



**KARNATAK UNIVERSITY, DHARWAD**  
**ACADEMIC (S&T) SECTION**  
**ಕರ್ನಾಟಕ ವಿಶ್ವವಿದ್ಯಾಲಯ, ಧಾರವಾಡ**

ವಿದ್ಯಾಮಂಡಳ (ಎಸ್&ಟಿ) ವಿಭಾಗ

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ಸಂ. KU/Aca(S&T)/OS-Gen/2026-27/ 1208

ದಿನಾಂಕ: 1 AUG 2026

**ಅಧಿಸೂಚನೆ**

ವಿಷಯ: 2026-27ನೇ ಶೈಕ್ಷಣಿಕ ಸಾಲಿನಿಂದ ಸ್ನಾತಕ ಪದವಿ-1, 2ನೇ ಹಾಗೂ 5, 6ನೇ ಸೆಮಿಸ್ಟರ್‌ಗಳಿಗೆ ಪಠ್ಯಕ್ರಮ ಅಳವಡಿಸುವ ಕುರಿತು.

ಉಲ್ಲೇಖ: ೧. ಎಲ್ಲ ನಿಖರತೆಗಳ ಸಭೆ ಜರುಗಿದ ದಿನಾಂಕ: 23 / 24 / 25.07.2026.

೨. ಮಾನ್ಯ ಕುಲಪತಿಗಳ ಅನುಮೋದನೆ ದಿನಾಂಕ: 20-07-2026.

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ಮೇಲ್ಕಾಣಿಸಿದ ವಿಷಯ ಹಾಗೂ ಉಲ್ಲೇಖಗಳಿಗೆ ಸಂಬಂಧಿಸಿದಂತೆ, ಮಾನ್ಯ ಕುಲಪತಿಗಳ ಅನುಮೋದನೆ ಮೇರೆಗೆ, ಸನ್ 2026-27ನೇ ಶೈಕ್ಷಣಿಕ ಸಾಲಿನಿಂದ ಕರ್ನಾಟಕ ವಿಶ್ವವಿದ್ಯಾಲಯದ ಸ್ನಾತಕ ಪದವಿಗಳಾದ BA / B.Com / B.Sc / BCA / BBA ಮತ್ತು ಇನ್ನಿತರ ವೃತ್ತಿಯೇತರ / ವೃತ್ತಿಪರ ಕೋರ್ಸುಗಳಿಗೆ 1 ಮತ್ತು 2ನೇ ಸೆಮಿಸ್ಟರ್ ಹಾಗೂ 2026-27ನೇ ಸಾಲಿಗೆ 5 ಮತ್ತು 6ನೇ ಸೆಮಿಸ್ಟರ್‌ಗಳಿಗೆ ಪಠ್ಯಕ್ರಮವನ್ನು ಅಳವಡಿಸುವ ಕುರಿತು ಮುಂಬರುವ ವಿದ್ಯಾವಿಷಯ ಪರಿಷತ್ ಸಭೆಯ ಸ್ಥಿರೀಕರಣಕ್ಕೊಳಪಟ್ಟು (Pending approval of Academic Council) ಅಧಿಸೂಚನೆ ಪ್ರಕಟಿಸಿದೆ. ಅದರಂತೆ, ಪಠ್ಯಕ್ರಮವನ್ನು ಕ.ವಿ.ವಿ. ಅಂತರ್ಜಾಲ [www.kud.ac.in](http://www.kud.ac.in) ದಲ್ಲಿ ಬಿತ್ತರಿಸಲಾಗಿದೆ. ಸದರ ಪಠ್ಯಕ್ರಮವನ್ನು ಕ.ವಿ.ವಿ. ಅಂತರ್ಜಾಲದಿಂದ ಡೌನ್‌ಲೋಡ್ ಮಾಡಿಕೊಳ್ಳಲು ಸೂಚಿಸುತ್ತ ವಿದ್ಯಾರ್ಥಿಗಳು ಹಾಗೂ ಸಂಬಂಧಿಸಿದ ಎಲ್ಲ ಬೋಧಕರ ಗಮನಕ್ಕೆ ತಂದು ಅದರಂತೆ ಕಾರ್ಯ ಪ್ರವೃತ್ತಿರಾಗಲು ಸೂಚಿಸಲಾಗಿದೆ.

ಅಡಕ: ಮೇಲಿನಂತೆ

*N. Ramiah*  
ಕುಲಸಚಿವರು 31/7/26

ಗೆ,  
ಕರ್ನಾಟಕ ವಿಶ್ವವಿದ್ಯಾಲಯದ ವ್ಯಾಪ್ತಿಯಲ್ಲಿ ಬರುವ ಎಲ್ಲ ಅಧೀನ ಹಾಗೂ ಸಂಬಂಧ ಪರಿಹಾರವಿದ್ಯಾಲಯಗಳ ಪ್ರಾಚಾರ್ಯರುಗಳಿಗೆ (ಕ.ವಿ.ವಿ. ಅಂತರ್ಜಾಲ ಹಾಗೂ ಮಿಂಚಂಚೆ ಮೂಲಕ ಬಿತ್ತರಿಸಲಾಗುವುದು)

ಪ್ರತಿ:

1. ನೋಡಲ್ ಅಧಿಕಾರಿಗಳು, UUCMS ಘಟಕ, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
2. ನಿರ್ದೇಶಕರು, ಐ.ಟಿ. ಶಾಖೆ, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
3. ಕುಲಪತಿಗಳ ಆಪ್ತ ಕಾರ್ಯದರ್ಶಿಗಳು, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
4. ಕುಲಸಚಿವರ ಆಪ್ತ ಕಾರ್ಯದರ್ಶಿಗಳು, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
5. ಕುಲಸಚಿವರು (ಮೇಲ್ವಿಚಾರಣೆ) ಆಪ್ತ ಕಾರ್ಯದರ್ಶಿಗಳು, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
6. ಅಧೀಕ್ಷಕರು, ಪ್ರಶ್ನೆ ಪತ್ರಿಕೆ / ಗೌಪ್ಯ / ಜಿ.ಎ.ಡಿ. / ವಿದ್ಯಾಂಡಳ (ಪಿ.ಜಿ.ಪಿ.ಎಚ್.ಡಿ) ವಿಭಾಗ, ಸಂಬಂಧಿಸಿದ ಕೋರ್ಸುಗಳ ವಿಭಾಗಗಳು ಪರೀಕ್ಷಾ ವಿಭಾಗ, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.



# **KARNATAK UNIVERSITY, DHARWAD**

**Programme: B.Sc.**

**Course: MATHEMATICS**

**SYLLABUS FOR I & II SEMESTER**

**(DSC and SEC)**

**UNDER**

**Revised National Education Policy (R-NEP) 2024**

**Revised Credit System (RC-NEP) 2026**

**Effect from 2026-27 onwards**

# KARNATAK UNIVERSITY, DHARWAD

Programme: B.Sc.

Course: MATHEMATICS

Effect from 2026-27 onwards

| Semester               | Type of Course    | Theory /Practical | Course Code      | Course Title                                 | Instruction Hours/week | Total Hours of syllabus/Sem | Duration of Sem-End Exam | Marks                |                      |       | Credits   |
|------------------------|-------------------|-------------------|------------------|----------------------------------------------|------------------------|-----------------------------|--------------------------|----------------------|----------------------|-------|-----------|
|                        |                   |                   |                  |                                              |                        |                             |                          | Formative Assessment | Summative Assessment | Total |           |
| <b>I</b>               | <b>DSC-1 (T1)</b> | <b>Theory</b>     | <b>CC1MAT1T1</b> | Algebra-I and Calculus-I                     | 04                     | 60                          | 03                       | 20                   | 80                   | 100   | <b>03</b> |
|                        | <b>DSC-1 (P1)</b> | <b>Practical</b>  | <b>CC1MAT1P1</b> | Practical on Algebra-I and Calculus-I        | 04                     | 56                          | 03                       | 10                   | 40                   | 50    | <b>02</b> |
| <b>II</b>              | <b>DSC-2 (T2)</b> | <b>Theory</b>     | <b>CC2MAT1T1</b> | Real Analysis-I and Calculus-II              | 04                     | 60                          | 03                       | 20                   | 80                   | 100   | <b>03</b> |
|                        | <b>DSC-2 (P2)</b> | <b>Practical</b>  | <b>CC2MAT1P1</b> | Practical on Real Analysis-I and Calculus-II | 04                     | 56                          | 03                       | 10                   | 40                   | 50    | <b>02</b> |
| <b>*II/<br/>III/IV</b> | <b>SEC</b>        | <b>Practical</b>  | <b>CC0MAT6P1</b> | Programming with GNU Octave                  | 04                     | 56                          | 03                       | 10                   | 40                   | 50    | <b>02</b> |

\*Student shall study this Skill Enhancement Course (SEC) either in 2<sup>nd</sup>/3<sup>rd</sup>/4<sup>th</sup> but not in all the semester.

# **KARNATAK UNIVERSITY, DHARWAD**

## **Programme: B.Sc. Course: MATHEMATICS**

### **Programme Specific Outcomes (PSOs):**

On completion of the 03-year Degree in Mathematics, students will be able to:

|             |                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PSO1</b> | Demonstrate a sound understanding of the fundamental concepts, principles, and techniques in algebra, calculus, differential equations, real and complex analysis, linear algebra, geometry and topology |
| <b>PSO2</b> | Apply mathematical methods and computational techniques to solve theoretical and practical problems arising in Science, Engineering, Economics, and Related fields                                       |
| <b>PSO3</b> | Develop logical reasoning, analytic thinking, and problem solving skills through proof techniques, quantitative analysis, mathematical modeling to solve interdisciplinary problems                      |
| <b>PSO4</b> | Use appropriate mathematical software and computational tools to analyze data, perform numerical computations, and visualize mathematical tools                                                          |
| <b>PSO5</b> | Communicate mathematical ideas, arguments, and solutions effectively in both written and oral forms while working independently and collaboratively in academia and industry                             |
| <b>PSO6</b> | Pursue higher studies, research, or professional careers by applying mathematical knowledge with ethical responsibility and lifelong learning skills                                                     |

**B.Sc. Semester–I**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Course (DSC)**  
**Course Title: Algebra-I and Calculus–I**  
**Course Code: CC1MAT1T1(Theory)**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC–1 (T1)     | Theory            | 03      | 04                         | 60 Hrs.                                    | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** After completion of course (Theory), students will be able to:

**CO1:** Learn to solve the system of homogeneous and non-homogeneous linear equations in  $n$  variables by using concept of rank of matrix, find eigenvalues and eigenvectors.

**CO2:** Understand the concepts of continuity.

**CO3:** Solve examples involving indeterminate forms.

**CO4:** Evaluate the radius of curvature in Cartesian and parametric forms.

**CO5:** Learn geometrical representation of mean value theorem and Rolle's theorem.

**CO6:** Identify and apply the Leibnitz theorem, intermediate value theorem and obtain Maclaurin's series expansions.

| Unit    | Title: Algebra-I and Calculus–I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 60Hours /Sem |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Unit-I  | <b>Matrices:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 15Hrs.       |
|         | Rank of a matrix based on row reduction to echelon form, Reduction to normal form, Solution of system of linear equations: Criterion for existence of non-trivial solutions of homogeneous system of linear equations, Solution of system of non-linear equations: Criterion for existence of non-trivial solutions of system of non-homogeneous linear equations, Characteristic equation of matrices, Eigenvalues and Eigenvectors of square matrices, Cayley-Hamilton theorem, Inverse of matrices by Cayley-Hamilton theorem. |              |
| Unit-II | <b>Limits and Continuity:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 15Hrs.       |
|         | Definition of limit and continuity of a function in $\varepsilon$ - $\delta$ form. Algebra of limits (with proof) and continuity, Definition of boundedness of continuous function, Properties of continuous function, Intermediate value theorem and its examples, Definition and problems on differentiability of a function, Indeterminate forms, Evaluation of limits using L-Hopital's Rule.                                                                                                                                 |              |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Unit-III | <b>Polar Coordinates:</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>15Hrs.</b> |
|          | Polar coordinates, Angle between the radius vector and tangent, Angle of intersection of curves (polar forms), Length of perpendicular from pole to the tangent, Pedal equations, Derivative of an arc length in Cartesian, parametric and polar forms, Concavity, Convexity and points of inflexion, Curvature of plane curve, Radius of curvature (Cartesian and Parametric forms), Centre of curvature of plane curves (Cartesian form).          |               |
| Unit-IV  | <b>Successive Differentiation and Mean Value Theorems:</b>                                                                                                                                                                                                                                                                                                                                                                                           | <b>15Hrs.</b> |
|          | The $n^{\text{th}}$ derivatives of standard functions: $e^{ax+b}$ , $(ax+b)^n$ , $\log(ax+b)$ , $\sin(ax+b)$ , $\cos(ax+b)$ , $e^{ax}\sin(bx+c)$ , $e^{ax}\cos(bx+c)$ with examples, Leibnitz theorem and its applications. Rolle's theorem, Lagrange's mean value theorem, Cauchy's mean value theorem and related problems. Taylor's theorem with Schlomitch and Roche form of remainder (statement only) Taylor's series, Maclaurin's expansions. |               |

### Recommended Books:

1. Theory of Matrices, B.S. Vatsa and S. Vatsa, New Age International(P) Ltd.,2010.
2. Matrices, A.R.Vasista and A.K.Vasista, Krishna Prakashan Media (P) Ltd.,2013.
3. Principles of Mathematical Analysis, W. Rudin, McGraw Hill Edu.,2023.
4. Mathematical Analysis, S.C.Malik and Savita Arora, New Age International(P)Ltd., 2021.
5. Introduction to Real Analysis, S.K.Mapa, Leveant Book Publishers,2022.
6. Differential Calculus, Shanti Narayan and P.K.Mittal,S.Chand & Co.,2022.
7. The Elements of Calculus, D.C. Pavate and G.V.Bhagwat, Popular Prakashan,1956.
8. Schaum's Outlines of Calculus, E.Mendelson,McGrawHill,2021.
9. A Course in B.Sc. Mathematics-I,S.S. Bhoosnurmath, C.S.Salimath and V.S.Shetiya EBPB, 2015.
10. Objective Mathematics – Differential Calculus and Integral Calculus, K. B. Pandey, A. K. Singh and V. Yadav, Pragati Prakashan, 2020.
11. A Course in B.Sc. Mathematics-I,C.S.Salimath,EBPB,2020.

| Formative Assessment for Theory               |                |
|-----------------------------------------------|----------------|
| Assessment Occasion / Type                    | Marks          |
| Internal Assessment Test-1                    | 05             |
| Internal Assessment Test-2                    | 05             |
| Assignment                                    | 10             |
| <b>Total</b>                                  | <b>20Marks</b> |
| <b>Formative Assessment as per guidelines</b> |                |

# **B.Sc. Semester–I**

## **Subject: MATHEMATICS**

### **Discipline Specific Core Course (DSC)**

#### **Course Title: Practical on Algebra–I and Calculus–I**

#### **Course Code: CC1MAT1P1 (Practical)**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC–1 (P1)     | Practical         | 02      | 04                         | 56 Hrs.                                    | 03 Hrs.                  | 10                         | 40                         | 50          |

**Course Outcomes (COs):** After completion of course (Practical), students will be able to:

**CO1:** Learn Scilab software, which is free and open source software for computer programming.

**CO2:** Solve problem on algebra and calculus using Scilab software.

**CO3:** Acquire knowledge of applications of algebra and calculus through Scilab software.

**CO4:** Calculate rank and row reduced echelon forms.

**CO5:** Evaluate the radius and Centre of curvatures in Cartesian form.

**CO6:** Obtain the Maclaurin's series expansions.

### **List of Experiments**

**[Each will have 4hrs./Week (Minimum 12 experiments)]**

1. Program to compute addition and subtraction of matrices.
2. Program to compute multiplication of matrices.
3. Program to compute rank of matrix and row reduced echelon form.
4. Program to check whether the given function is continuous at a point or not.
5. Program to solve examples based on Intermediate value theorem.
6. Program to find a length of perpendicular from pole to the tangent.
7. Program to find the angle between the radius vector and tangent.
8. Program to find the radius of curvatures of the given curves (Cartesian form).
9. Program to find the coordinates of centre of curvatures of the given curves (Cartesian form)
10. Program to find the  $n^{\text{th}}$  derivative of  $e^{ax+b}$ . and trigonometric functions.
11. Program to find the  $n^{\text{th}}$  derivative of  $e^{ax}\sin(bx+c)$  and  $e^{ax}\cos(bx+c)$ .
12. Program to find the Maclaurin's expansions of the given functions.

### Recommended Books:

1. Scilab by Example, M. Affouf, Create Space Independent Publishing Platform, 2012.
2. Scilab (A free software to Matlab), H. Ramchandran and A.S. Nair, S. Chand &Co., 2018.
3. Scilab–A Beginner’s Approach, A.K. Verma, Cengage Learning India Pvt.Ltd., 2018.
4. Programming in Scilab, R.Goyal and M.Dhingra, Narosa Publishing, 2019.
5. Computing in Scilab, C. Jain, Cambridge University Press, 2022.
6. University Algebra, N.S. Gopalakrishnan, New Age International (P)Limited,2018.
7. Theory of Matrices, B.S. Vatsa and S. Vatsa, New Age International(P)Ltd.,2010.
8. Matrices, A.R. Vasista and A.K. Vasista, Krishna Prakashan Media (P)Ltd.,2013.
9. Principles of Mathematical Analysis, W. Rudin, Mc Graw Hill Edu., 2023.
10. Mathematical Analysis, S.C. Malik and Savita Arora, New Age International(P)Ltd., 2021.

### General Instructions

**Software to be used: SCILAB**

### **Scheme of Practical Examination (Distribution of Marks)**

| Formative Assessment for Practical              |                |
|-------------------------------------------------|----------------|
| Assessment Occasion/Type                        | Marks          |
| Program writing and problem solving (1 Program) | 03             |
| Program Execution                               | 03             |
| Viva                                            | 02             |
| Journal                                         | 02             |
| <b>Total</b>                                    | <b>10Marks</b> |
| <i>Formative Assessment as per guidelines.</i>  |                |

| Summative Assessment for Practical               |                |
|--------------------------------------------------|----------------|
| Assessment Occasion/Type                         | Marks          |
| Program writing and problem solving (2 Programs) | (5+5)*2=20     |
| Program Execution (Any 1 Program)                | 10             |
| Viva                                             | 05             |
| Journal                                          | 05             |
| <b>Total</b>                                     | <b>40Marks</b> |
| <i>Summative Assessment as per guidelines.</i>   |                |

**B.Sc. Semester-II**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Course (DSC)**  
**Course Title: Real Analysis-I and Calculus-II**  
**Course Code: CC2MAT1T1 (Theory)**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC-2 (T2)     | Theory            | 03      | 04                         | 60 Hrs.                                    | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** After completion of course (Theory), students will be able to:

**CO 1:** Learn countable and uncountable sets.

**CO 2:** Understand the concepts of open set, closed set, etc.

**CO 3:** Study the applications of definite integrals to areas, volumes and surface of revolution.

**CO 4:** Learn maxima and minima of the functions of two variables.

**CO 5:** Find the extreme values of the functions.

**CO 6:** Understand the concepts of gradient, divergence and curl.

| Unit            | Title: Real Analysis-I and Calculus-II                                                                                                                                                                                                                                                                                                                                                                                                        | 60Hours / Sem |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Unit-I</b>   | <b>Introduction to Real Analysis:</b>                                                                                                                                                                                                                                                                                                                                                                                                         | 15Hrs.        |
|                 | Real line, Field and order axioms, Intervals, Bounded and unbounded sets, Supremum and infimum of a set, Completeness property of $\mathbb{R}$ , Archimedean property of $\mathbb{R}$ . Absolute value of $\mathbb{R}$ – Standard theorems, Neighbourhood of a point, Interior point of a set, Open set, Closed set, Limit points of a set, Bolzano-Weierstrass theorem (statement only), Countable and uncountable sets - Standard theorems. |               |
| <b>Unit-II</b>  | <b>Integral Calculus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | 15Hrs.        |
|                 | Reduction Formulae, application of definite integrals to areas, volumes and Surface of revolution, Length of plane curves.                                                                                                                                                                                                                                                                                                                    |               |
| <b>Unit-III</b> | <b>Multivariate Calculus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | 15Hrs.        |
|                 | Functions of two or more variables, Partial derivatives of implicit and composite functions, Euler's theorem and its extension, Total differentials, Jacobians and standard properties and illustrative examples, Taylor's and Maclaurin's series for functions of two variables (Statement Only), Maxima-Minima of functions of two variables, Lagrange's method of undetermined multipliers.                                                |               |

|                |                                                                                                                                                                                                                                                                                                                                                       |        |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <b>Unit–IV</b> | <b>Vector Calculus:</b>                                                                                                                                                                                                                                                                                                                               | 15Hrs. |
|                | Dot and cross product of vectors. Ordinary derivatives of vectors. Continuity and differentiability of a vector function. Derivatives of sum, dot product. Cross product and triple products of vectors. Differential of vectors. The vector differential operator $\nabla$ . The gradient, Divergence and curl. Solenoidal and Irrotational vectors. |        |

### Recommended Books:

1. Real Analysis, N.P.Bali (Golden Maths Series), New Age International Publishers, 2023.
2. Real Analysis, M.L.Khanna and S.K.Pundir, Jai Prakash Nath & Co. Meerut, 2014.
3. Introduction to Real Analysis, R.G.Bartle and D.R.Sherbert, Wiley, 2021.
4. Integral Calculus, Shanti Narayan and P.K. Mittal, S. Chand and Co. Pvt. Ltd., 2005.
5. Integral Calculus, S.K. Pundir and B. Singh, Pragati Prakashan, 2020.
6. Differential Calculus, Shanti Narayan and P. K. Mittal, S. Chand & Company, New Delhi, 2022.
7. Schaum's Outlines of Calculus, E. Mendelson, Mc GrawHill, 2021.
8. Vector Calculus, J.N. Sharma and A.R. Vasishtha, Krishna Prakashan, 2020.
9. A Text book of Vector Calculus, Shanti Narayan and P.K. Mittal, Visionias, 2023.
10. Vector Calculus, R.K. Pandey, Oxford Publications, 2012.
11. A Course in B.Sc. Mathematics-II, S. S. Bhoosnurmath, C. S. Salimath and V. S. Shetiya, EBPB, 2016.
12. A Text Book of B.Sc. Mathematics, G.K. Ranganath, S. Chand Publications, 2009.
13. Mathematical Methods, S.K.Pundir and B.Singh, Pragati Prakashan, 2020.
14. A Course in B.Sc. Mathematics-II, C.S. Salimath, EBPB, 2021.
15. Objective Mathematics – Differential Calculus and Integral Calculus, K. B. Pandey, A. K. Singh and V. Yadav, Pragati Prakashan, 2020.

| <b>Formative Assessment for Theory</b>        |                |
|-----------------------------------------------|----------------|
| <b>Assessment Occasion / Type</b>             | <b>Marks</b>   |
| Internal Assessment Test-1                    | 05             |
| Internal Assessment Test-2                    | 05             |
| Assignment                                    | 10             |
| <b>Total</b>                                  | <b>20Marks</b> |
| <b>Formative Assessment as per guidelines</b> |                |

## B.Sc. Semester–II

### Subject: MATHEMATICS

#### Discipline Specific Core Course (DSC)

#### Course Title: Practical on Real Analysis-I and Calculus–II

#### Course Code: CC2MAT1P1 (Practical)

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC–2 (P2)     | Practical         | 02      | 04                         | 56Hrs.                                     | 03Hrs.                   | 10                         | 40                         | 50          |

**Course Outcomes (COs):** After completion of course (Practical), students will be able to:

**CO 1:** Learn Scilab software, which is free and open source software for computer programming.

**CO 2:** Acquire knowledge of applications of real analysis.

**CO 3:** Apply reduction formulae to evaluate integrals.

**CO 4:** Verify Euler’s theorem and its extension for homogeneous functions.

**CO 5:** Analyse the dot and cross products of vectors.

#### List of Experiments

[Each will have 4 Hrs./Week (Minimum 12 experiments)]

1. Program to find infimum and supremum of a set by suitable examples.
2. Program to check whether a set is bounded or unbounded.
3. Program to compute the absolute value of a real number.
4. Program on reduction formulas for  $\int \sin^n x dx$  and  $\int \cos^n x dx$  (for all positive values of  $n$ ) with suitable examples.
5. Program on reduction formulas  $\int_0^{\pi/2} \sin^n x dx$  and  $\int_0^{\pi/2} \cos^n x dx$  (for all positive odd and even integral values of  $n$ ) with suitable examples.
6. Program on reduction formulas for  $\int \tan^n x dx$  and  $\int \cot^n x dx$  (for all positive values of  $n$ ) with suitable examples.
7. Program to find First order Partial derivatives of function  $f(x, y)$
8. Program to verify the Euler’s theorem and its extension for homogeneous functions.
9. Program to compute the Jacobian of two function w.r.t. two variables.
10. Program to find the dot product, Cross product and triple products of vectors.

11. Program to verify whether a vector field is Solenoidal or irrotational.
12. Program to find gradient, divergence and curl.

### Recommended Books:

1. Scilab by Example, M.Affouf, Create Space Independent Publishing Platform, 2012.
2. Scilab (A free software to Matlab), H. Ramchandran and A.S.Nair, S.Chand & Co., 2018.
3. Scilab—A Beginner's Approach, A.K.Verma, Cengage Learning India Pvt. Ltd., 2018.
4. Programming in Scilab, R.Goyal and M. Dhingra, Narosa Publishing, 2019.
5. Computing in Scilab, C. Jain, Cambridge University Press, 2022.
6. Vector Calculus, J.N.Sharma and A.R.Vasishtha, Krishna Prakashan, 2020.
7. Integral Calculus, Shanti Narayan and P.K. Mittal, S. Chand and Co.Pvt. Ltd., 2005.
8. Differential Calculus, Shanti Narayan and P.K.Mittal, S.Chand & Co., New Delhi, 2022.
9. Schaum's Outlines of Calculus, E. Mendelson, Mc Graw Hill, 2021.
10. A Text Book of B.Sc. Mathematics, G.K.Ranganath, S.Chand Publications, 2009.

### General Instructions

**Software to be used: SCILAB**

### **Scheme of Practical Examination (Distribution of Marks)**

| Formative Assessment for Practical              |                |
|-------------------------------------------------|----------------|
| Assessment Occasion/Type                        | Marks          |
| Program writing and problem solving (1 Program) | 03             |
| Program Execution                               | 03             |
| Viva                                            | 02             |
| Journal                                         | 02             |
| <b>Total</b>                                    | <b>10Marks</b> |
| <i>Formative Assessment as per guidelines.</i>  |                |

| Summative Assessment for Practical               |                 |
|--------------------------------------------------|-----------------|
| Assessment Occasion/Type                         | Marks           |
| Program writing and problem solving (2 Programs) | (5+5)*2=20      |
| Program Execution (Any 1 Program)                | 10              |
| Viva                                             | 05              |
| Journal                                          | 05              |
| <b>Total</b>                                     | <b>40 Marks</b> |
| <i>Summative Assessment as per guidelines</i>    |                 |

**B.Sc. Semester –II/III/IV**  
**Subject: MATHEMATICS**  
**SKILL ENHANCEMENT COURSE (SEC)**  
**(Student shall study this SEC in any one of the Semesters**  
**either in II or III or IV semester and College shall decide to**  
**allot the students)**  
**Course Title: Programming with GNU Octave Course**  
**Code: CC0MAT6P1 (Practical)**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| SEC            | Practical         | 02      | 04                         | 56Hrs.                                     | 03Hrs.                   | 10                         | 40                         | 50          |

**Course Outcomes (COs):** After completion of course(Practical),students will be able to:

CO1: Learn GNU Octave Software (FOSS) for computer programming.

CO2: Solve problems on Algebra, Calculus by using GNU Octave.

CO3: Verify mathematical concepts and theorems using GNU Octave.

CO4: Apply problem-solving skills using GNU Octave.

**List of Experiments**

**[Each will have 4Hrs/Week (Minimum 12 experiments)]**

1. Program to compute addition and subtraction of two matrices.
2. Program to compute multiplication of two matrices.
3. Program to check whether the given function is continuous at a point or not.
4. Program to find the angle between the radius vector and tangent.
5. Program to find the radius of curvatures of the given curves
6. Program to find the  $n^{\text{th}}$  derivative of  $e^{ax}$  and trigonometric functions.
7. Program on reduction formulas  $\int_0^{\pi/2} \sin^n x dx$  and  $\int_0^{\pi/2} \cos^n x dx$  (for all positive odd and even integral values of  $n$ ) with suitable examples.
8. Program to solve examples using Euler's theorem.
9. Program to find First order Partial derivatives of function  $f(x, y)$
10. Program to compute the Jacobin of two function w.r.t. two variables.
11. Program to find the dot product, Cross product and triple products of vectors.
12. Program to verify whether a vector field Solenoidal or irrotational.

## Recommended Books:

1. Introduction to Octave, N. Sandeep, APress, 2017.
2. Scientific Computing with Matlab and Octave, A. Quarteroni, Springer, 2010.
3. Essential Matlab and Octave, J. Rogel-Salazar, CRC Press, 2018.
4. GNU Octave by Example, A. Pajankar, APress, 2020.
5. Octave and Matlab for Engineering Applications, A. Stahel, Springer, 2022.
6. Differential Calculus, Shanti Narayan and P. K. Mittal, S. Chand & Company, New Delhi, 2022.
7. Modern Algebra, S. Singh and Q. Zameeruddin, Vikas Publishing House Pvt. Ltd., 2006.
8. Vector Calculus, J. N. Sharma and A. R. Vasishtha, Krishna Prakashan, 2020.

### General Instructions

**Software to be used: GNU OCTAVE**

#### **Scheme of Practical Examination (Distribution of Marks)**

| <b>Formative Assessment for Practical</b>       |                 |
|-------------------------------------------------|-----------------|
| <b>Assessment Occasion/Type</b>                 | <b>Marks</b>    |
| Program writing and problem solving (1 Program) | 03              |
| Program Execution                               | 03              |
| Viva                                            | 02              |
| Journal                                         | 02              |
| <b>Total</b>                                    | <b>10 Marks</b> |
| <i>Formative Assessment as per guidelines.</i>  |                 |

| <b>Summative Assessment for Practical</b>        |                 |
|--------------------------------------------------|-----------------|
| <b>Assessment Occasion/Type</b>                  | <b>Marks</b>    |
| Program writing and problem solving (2 Programs) | $(5+5)*2=20$    |
| Program Execution (Any 1 Program)                | 10              |
| Viva                                             | 05              |
| Journal                                          | 05              |
| <b>Total</b>                                     | <b>40 Marks</b> |
| <i>Summative Assessment as per guidelines.</i>   |                 |

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## **B.Sc. Programme: 2026-27**

GENERAL PATTERN OF THEORY QUESTION PAPER FOR DSC  
(80 Marks for Semester-End Examination with 03 Hrs duration)

### **Part-A**

1. Question Numbers 1-10 carry 2 marks each. :20 Marks

### **Part-B**

2. Question Numbers 11-18 carry 05 Marks each. Answer any 06 questions :30 Marks

### **Part-C**

3. Question Numbers 19-22 carry 10 Marks each. Answer any 03 questions :30 Marks

(Minimum 1 question from each unit and 10 marks question may have Sub-questions for 7+3 or 6+4 or 5+5 if necessary)

**Total: 80 Marks**

**Note: Equal weightage shall be given to each unit based on number of hours prescribed.**

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**KARNATAK UNIVERSITY, DHARWAD**

**ACADEMIC (S&T) SECTION**  
**ಕರ್ನಾಟಕ ವಿಶ್ವವಿದ್ಯಾಲಯ, ಧಾರವಾಡ**

**ವಿದ್ಯಾಕುಲ (ಎಸ್&ಟಿ) ವಿಭಾಗ**

Tele: 0836-2215224  
e-mail: academic.st@kud.ac.in

website: kud.ac.in

Private Nagar, Dharwad-580003  
ಪಿ.ಎಚ್. ನಗರ, ಧಾರವಾಡ-580003

ಸಂ. KU/Aca(S&T)/OS-Gen/2026-27/ 1208

ದಿನಾಂಕ: 1 AUG 2026

**ಅಧಿಸೂಚನೆ**

ವಿಷಯ: 2026-27ನೇ ಶೈಕ್ಷಣಿಕ ಸಾಲಿನಿಂದ ಸ್ನಾತಕ ಪದವಿ-1, 2ನೇ ಹಾಗೂ 5, 6ನೇ ಸೆಮಿಸ್ಟರ್‌ಗಳಿಗೆ ಪಠ್ಯಕ್ರಮ ಅಳವಡಿಸುವ ಕುರಿತು.

ಉಲ್ಲೇಖ: ೧. ಎಲ್ಲ ನಿಖಾಯಗಳ ಸಭೆ ಜರುಗಿದ ದಿನಾಂಕ: 23 / 24 / 25.07.2026.

೨. ಮಾನ್ಯ ಕುಲಪತಿಗಳ ಅನುಮೋದನೆ ದಿನಾಂಕ: 20-07-2026.

\*\*\*\*\*

ಮೇಲ್ಕಾಣಿಸಿದ ವಿಷಯ ಹಾಗೂ ಉಲ್ಲೇಖಗಳಿಗೆ ಸಂಬಂಧಿಸಿದಂತೆ, ಮಾನ್ಯ ಕುಲಪತಿಗಳ ಅನುಮೋದನೆ ಮೇರೆಗೆ, ಸನ್ 2026-27ನೇ ಶೈಕ್ಷಣಿಕ ಸಾಲಿನಿಂದ ಕರ್ನಾಟಕ ವಿಶ್ವವಿದ್ಯಾಲಯದ ಸ್ನಾತಕ ಪದವಿಗಳಾದ BA / B.Com / B.Sc / BCA / BBA ಮತ್ತು ಇನ್ನಿತರ ವೃತ್ತಿಯೇತರ / ವೃತ್ತಿಪರ ಕೋರ್ಸುಗಳಿಗೆ 1 ಮತ್ತು 2ನೇ ಸೆಮಿಸ್ಟರ್ ಹಾಗೂ 2026-27ನೇ ಸಾಲಿಗೆ 5 ಮತ್ತು 6ನೇ ಸೆಮಿಸ್ಟರ್‌ಗಳಿಗೆ ಪಠ್ಯಕ್ರಮವನ್ನು ಅಳವಡಿಸುವ ಕುರಿತು ಮುಂಬರುವ ವಿದ್ಯಾವಿಷಯಕ ಪರಿಷತ್ ಸಭೆಯ ಸ್ನೇಹಿತರಣಕ್ಕೊಳಪಟ್ಟು (Pending approval of Academic Council) ಅಧಿಸೂಚನೆ ಪ್ರಕಟಿಸಿದೆ. ಅದರಂತೆ, ಪಠ್ಯಕ್ರಮವನ್ನು ಕ.ವಿ.ವಿ. ಅಂತರ್ಜಾಲ [www.kud.ac.in](http://www.kud.ac.in) ದಲ್ಲಿ ಬಿತ್ತರಿಸಲಾಗಿದೆ. ಸದರ ಪಠ್ಯಕ್ರಮವನ್ನು ಕ.ವಿ.ವಿ. ಅಂತರ್ಜಾಲದಿಂದ ಡೌನ್‌ಲೋಡ್ ಮಾಡಿಕೊಳ್ಳಲು ಸೂಚಿಸುತ್ತ ವಿದ್ಯಾರ್ಥಿಗಳು ಹಾಗೂ ಸಂಬಂಧಿಸಿದ ಎಲ್ಲ ಬೋಧಕರ ಗಮನಕ್ಕೆ ತಂದು ಅದರಂತೆ ಕಾರ್ಯ ಪ್ರವೃತ್ತಿರಾಗಲು ಸೂಚಿಸಲಾಗಿದೆ.

ಆಡಕ: ಮೇಲಿನಂತೆ

*N. Ramulu*  
ಕುಲಸಚಿವರು 31/7/26

ಗೆ,

ಕರ್ನಾಟಕ ವಿಶ್ವವಿದ್ಯಾಲಯದ ವ್ಯಾಪ್ತಿಯಲ್ಲಿ ಬರುವ ಎಲ್ಲ ಅಧೀನ ಹಾಗೂ ಸಂलग್ನ ಮಹಾವಿದ್ಯಾಲಯಗಳ ಪ್ರಾಚಾರ್ಯರುಗಳಿಗೆ, (ಕ.ವಿ.ವಿ. ಅಂತರ್ಜಾಲ ಹಾಗೂ ಮಿಂಚಂಚೆ ಮೂಲಕ ಬಿತ್ತರಿಸಲಾಗುವುದು)

ಪ್ರತಿ:

1. ನೋಡಲ್ ಅಧಿಕಾರಿಗಳು, UUCMS ಘಟಕ, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
2. ನಿರ್ದೇಶಕರು, ಐ.ಟಿ. ಶಾಖೆ, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
3. ಕುಲಪತಿಗಳ ಆಪ್ತ ಕಾರ್ಯದರ್ಶಿಗಳು, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
4. ಕುಲಸಚಿವರ ಆಪ್ತ ಕಾರ್ಯದರ್ಶಿಗಳು, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
5. ಕುಲಸಚಿವರು (ಮೌಲ್ಯಮಾಪನ) ಆಪ್ತ ಕಾರ್ಯದರ್ಶಿಗಳು, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.
6. ಅಧೀಕ್ಷಕರು, ಪ್ರಶ್ನೆ ಪತ್ರಿಕೆ / ಗೌಪ್ಯ / ಜಿ.ಎ.ಡಿ. / ವಿದ್ಯಾಂಧಳ (ಪಿ.ಜಿ.ಪಿ.ಎಚ್.ಡಿ) ವಿಭಾಗ, ಸಂಬಂಧಿಸಿದ ಕೋರ್ಸುಗಳ ವಿಭಾಗಗಳು ಪರೀಕ್ಷಾ ವಿಭಾಗ, ಕ.ವಿ.ವಿ. ಧಾರವಾಡ.



**KARNATAK UNIVERSITY, DHARWAD**

**Programme: B.Sc.**

**Course: MATHEMATICS**

**SYLLABUS FOR V & VI SEMESTER**

**(DSC, DSE, and SEC)**

**UNDER**

**Revised National Education Policy (R-NEP) 2024**

**Effect from 2026-27 onwards**

**KARNATAK UNIVERSITY, DHARWAD****Programme: B.Sc.****Course: MATHEMATICS****Effect from 2026-27 onwards**

| Semester              | Type of Course                               | Theory /Practical | Course Code                           | Course Title                                                                      | Instruction hours/week | Total hours of syllabus/Sem | Duration of Sem End Exam | Marks                |                      |       | Credits   |
|-----------------------|----------------------------------------------|-------------------|---------------------------------------|-----------------------------------------------------------------------------------|------------------------|-----------------------------|--------------------------|----------------------|----------------------|-------|-----------|
|                       |                                              |                   |                                       |                                                                                   |                        |                             |                          | Formative Assessment | Summative Assessment | Total |           |
| <b>*V</b>             | <b>DSC – 5 (T5)</b>                          | <b>Theory</b>     | <b>C5 MAT1T1</b>                      | Real Analysis – II                                                                | 04                     | 60                          | 03                       | 20                   | 80                   | 100   | <b>03</b> |
|                       | <b>DSC – 5 (P5)</b>                          | <b>Practical</b>  | <b>C5 MAT1P1</b>                      | Practical on Real Analysis – II                                                   | 04                     | 56                          | 03                       | 10                   | 40                   | 50    | <b>02</b> |
|                       | <b>DSE – 1 (T6)<br/>or<br/>DSE – 2 (T7)</b>  | <b>Theory</b>     | <b>C5 MAT2T1<br/>or<br/>C5 MAT2T2</b> | Algebra-III and Special Functions<br>or<br>Linear Algebra and Analytical Geometry | 04                     | 60                          | 03                       | 20                   | 80                   | 100   | <b>03</b> |
| <b>*VI</b>            | <b>DSC – 6 (T8)</b>                          | <b>Theory</b>     | <b>C6 MAT1T1</b>                      | Numerical Analysis                                                                | 04                     | 60                          | 03                       | 20                   | 80                   | 100   | <b>03</b> |
|                       | <b>DSC – 6 (P6)</b>                          | <b>Practical</b>  | <b>C6 MAT1P1</b>                      | Practical on Numerical Analysis                                                   | 04                     | 56                          | 03                       | 10                   | 40                   | 50    | <b>02</b> |
|                       | <b>DSE – 3 (T9)<br/>or<br/>DSE – 4 (T10)</b> | <b>Theory</b>     | <b>C6 MAT2T1<br/>or<br/>C6 MAT2T2</b> | Complex Analysis and Topology<br>or<br>Mechanics and Mathematical Modelling       | 04                     | 60                          | 03                       | 20                   | 80                   | 100   | <b>03</b> |
| <b>**IV/<br/>V/VI</b> | <b>SEC</b>                                   | <b>Practical</b>  | <b>C0 MAT6 P1</b>                     | Programming with GNU Octave                                                       | 04                     | 56                          | 03                       | 10                   | 40                   | 50    | <b>02</b> |
| <b>***VI</b>          | <b>DSE-5 (PR)<br/>or<br/>DSE-6 (IN)</b>      | <b>Practical</b>  | <b>C6 MAT7PR<br/>or<br/>C6 MAT7IN</b> | Project Work / Internship                                                         | 04                     | 56                          | 03                       | 10                   | 40                   | 50    | <b>02</b> |

\*Student shall study T5 & P5 compulsory and either T6 or T7 in 5<sup>th</sup> semester. Similarly, T8 & P6 compulsory and either T9 or T10 in 6<sup>th</sup> semester.

\*\* Student shall study this Skill Enhancement Course (SEC) either in 4<sup>th</sup> / 5<sup>th</sup> / 6<sup>th</sup> but not in all the semesters.

\*\*\* Student shall opt Project Work/ Internship in any one of the major courses.

**B.Sc. Semester – V**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Course (DSC)**  
**Course Title: Real Analysis – II**  
**Course Code: C5MAT1T1 (Theory)**  
**Compulsory Theory paper**

| Type of Course | Theory /Practical | credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC – 5 (T5)   | Theory            | 03      | 04                         | 60 Hrs.                                    | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** After completion of the course (Theory), students will be able to:

**CO1:** Understand the concepts of Riemann integration and integrability.

**CO2:** Apply techniques of definite integration to solve problems.

**CO3:** Analyze and evaluate improper integrals using convergence tests.

**CO4:** Analyze Beta and Gamma functions, their properties, relations, and applications.

| Unit              | Title: Real Analysis – II                                                                                                                                                                                                                                                                                                                                    | 60 Hours / Sem |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Unit - I</b>   | <b>Riemann Integration-I:</b>                                                                                                                                                                                                                                                                                                                                | 15 Hrs.        |
|                   | Definition and examples for partition of an interval, Refinement of a partition and common refinement, Upper and lower Darboux sums, Upper and lower integrals, Riemann integral, Criterion for integrability, Integrability of sum, difference, product, quotient and modulus of integrable functions. Integral as a limit of sum (Riemann sum) – Problems. |                |
| <b>Unit – II</b>  | <b>Riemann Integration-II</b>                                                                                                                                                                                                                                                                                                                                | 15 Hrs.        |
|                   | Integrability of continuous functions, Monotonic functions, Bounded functions with finite number of discontinuity, Fundamental theorem of Calculus – related problems, Change of variables, Integration by parts, First and second mean value theorems of integral calculus.                                                                                 |                |
| <b>Unit – III</b> | <b>Improper Integrals:</b>                                                                                                                                                                                                                                                                                                                                   | 15 Hrs.        |

|                  |                                                                                                                                                                                                                                              |         |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|                  | Improper integrals of the first, second and third kind with examples, Improper integral as the limit of proper integral, Comparison test Abel's test and Dirichlet's test for the convergence of the integral of a product of two functions. |         |
| <b>Unit - IV</b> | <b>Beta-Gamma Functions:</b>                                                                                                                                                                                                                 | 15 Hrs. |
|                  | Definitions, Properties and examples, Relations between Beta and Gamma functions, Standard theorems, Duplication formula and related problems.                                                                                               |         |

#### Recommended Books:

1. Principles of Real Analysis, S. C. Malik, New Age International Publishers, 2021.
2. Mathematical Analysis, S. C. Malik and Savita Arora, New Age International (P) Ltd., 2021.
3. Real Analysis, N. P. Bali (Golden Maths Series), New Age International Publishers, 2023.
4. Integral Calculus, Shanti Narayan and P. K. Mittal, S. Chand Publications, 2005.
5. Integral Calculus, A. R. Vasishtha, R. Dangwal and R. Sharma, Krishna Prakashan, 2020.
6. A Course in B.Sc. Mathematics-V, S. S. Bhoosnurmath, C. S. Salimath and V. S. Shetiya (EBPB), 2018.

| Formative Assessment for Theory               |                |
|-----------------------------------------------|----------------|
| Assessment Occasion / Type                    | Marks          |
| Internal Assessment Test-1                    | 05             |
| Internal Assessment Test-2                    | 05             |
| Assignment                                    | 10             |
| <b>Total</b>                                  | <b>20Marks</b> |
| <b>Formative Assessment as per guidelines</b> |                |

**B.Sc. Semester – V**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Course (DSC)**  
**Course Title: Practical on Real Analysis – II**  
**Course Code: C5MAT1P1 (Practical)**  
**Compulsory Practical**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC – 5 (P5)   | Practical         | 02      | 04                         | 56 Hrs.                                    | 03 Hrs.                  | 10                         | 40                         | 50          |

**Course Outcomes (COs):** After completion of course (Practical), students will be able to:

- CO1:** Learn **Python (FOSS)** for mathematical computing.
- CO2:** Apply Python to compute Riemann sums and test Riemann integrability.
- CO3:** Use Python to evaluate definite and improper integrals and verify their convergence.
- CO4:** Compute and verify Beta and Gamma functions using Python.

**List of Experiments**

**[Each will have 4 Hrs. / Week (Minimum 12 experiments)]**

1. Program to find upper and lower Riemann sums with respect to a given partition
2. Program to test Riemann Integrability.
3. Program to evaluate Riemann integral as a limit of sum.
4. Program to verify the First Mean Value Theorem of Integral Calculus.
5. Program to verify the Second Mean Value Theorem of Integral Calculus.
6. Program to verify the Fundamental Theorem of Calculus with suitable examples.
7. Program to verify the convergence of an improper integral of first kind.
8. Program to verify the convergence of an improper integral of second kind.
9. Program to illustrate Dirichlet's test and Abel's Test with suitable example.
10. Program to evaluate Beta and Gamma functions for different values.
11. Program to verify the Beta–Gamma relation with suitable examples.
12. Program to verify the recurrence relation and duplication formula of the Gamma function with suitable examples.

**Recommended Books:**

1. Automate the Boring Stuff with Python, A. Sweigart, No Starch Press, 2019.
2. Python Cookbook, D. Beazley and B. K. Jones, Shroff O'Reilly, 2013.
3. Basic Python Programming for Beginners, K. V. Rajkumar, M. Krishna, and J. Prakash, Bluerose Publishers Pvt. Ltd., 2021.
4. Python, J. Shovic and A. Simpson, For Dummies, 2024.
5. Learning Python, M. Lutz, Shroff O'Reilly, 2017.
6. Mathematical Analysis, S. C. Malik and Savita Arora, New Age International (P) Ltd., 2021.
7. Real Analysis, N. P. Bali (Golden Maths Series), New Age International Publishers, 2023.
8. Integral Calculus, Shanti Narayan and P. K. Mittal, S. Chand Publications, 2005.

**General Instructions****Software to be used: PYTHON****Scheme of Practical Examination (Distribution of Marks)**

| Formative Assessment for Practical              |                 |
|-------------------------------------------------|-----------------|
| Assessment Occasion /Type                       | Marks           |
| Program writing and problem solving (1 Program) | 03              |
| Program Execution                               | 03              |
| Viva                                            | 02              |
| Journal                                         | 02              |
| <b>Total</b>                                    | <b>10 Marks</b> |
| <b>Formative Assessment as per guidelines.</b>  |                 |

| Summative Assessment for Practical               |                 |
|--------------------------------------------------|-----------------|
| Assessment Occasion /Type                        | Marks           |
| Program writing and problem solving (2 Programs) | $(5+5)*2 = 20$  |
| Program Execution (Any 1 Program)                | 10              |
| Viva                                             | 05              |
| Journal                                          | 05              |
| <b>Total</b>                                     | <b>40 Marks</b> |
| <b>Summative Assessment as per guidelines.</b>   |                 |

**B.Sc. Semester – V**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Elective (DSE)**  
**Course Title: Algebra-III and Special Functions**  
**Course Code: C5MAT2T1 (Theory)**

**SPECIALIZATION – I**

(Student shall select DSE –1 (T6) or DSE-2 (T7) for 03 credits only)

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSE – 1 (T6)   | Theory            | 03      | 04                         | 60Hrs.                                     | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** At the end of the course, students will be able to:

**CO 1:** Understand the concepts of Rings, Integral Domains, Fields

**CO 2:** Become familiar with the concepts of principal, prime and maximal ideals

**CO 3:** Apply the properties of polynomial rings, divisibility, and irreducibility of polynomials.

**CO 4:** Solve differential equations using Power series and Frobenius methods.

**CO 5:** Analyze Bessel functions, their recurrence relations, and generating functions.

**CO 6:** Apply the properties, recurrence relations, generating functions, and orthogonality of Legendre polynomials in solving mathematical problems.

| Unit       | Title: Algebra-III                                                                                                                                                                                                                                                                                                                                                                    | 60 Hours / Sem |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Unit – I   | <b>Rings-I, Integral Domains, Fields</b>                                                                                                                                                                                                                                                                                                                                              | 15 Hrs.        |
|            | Definition, examples. Abelian Ring. Ring with unity and without unity. Finite and infinite rings (Only definitions). Units in a ring. Zero divisors in a ring. Ring without zero divisor. Properties of ring. Division ring. Integral domain and field definitions. Theorems (i) A finite integral domain is a field (ii) Every field is an integral domain                           |                |
|            | <b>Rings-II, Ideals</b>                                                                                                                                                                                                                                                                                                                                                               | 15 Hrs         |
|            | Subrings, Polynomial rings, Divisibility, Irreducible polynomials, Eisenstein Criterion of Irreducibility, Ideals: Definition, theorems, examples. Maximal Ideal, Prime Ideal.                                                                                                                                                                                                        | .              |
| Unit – III | <b>Bessel's Functions:</b>                                                                                                                                                                                                                                                                                                                                                            | 15 Hrs         |
|            | Solution to differential equation - Ordinary, singular and regular points of second order linear differential equation, Power series solution when $x=0$ is an ordinary point, Frobenius method. Bessel's differential equation - Definition and discussion of its solutions; Bessel's function $J_n(x)$ - Definition, various recurrence relations for Bessel function (derivation), | .              |

|                  |                                                                                                                                                                                                                                                                                       |        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                  | Generating function for $J_n(x)$ (derivation), value of $J_{1/2}(x)$ and expansions for $J_0$ and $J_1$ and related problems.                                                                                                                                                         |        |
| <b>Unit – IV</b> | <b>Legendre's Functions:</b>                                                                                                                                                                                                                                                          | 15 Hrs |
|                  | Legendre function- Discussion of solutions to Legendre's differential equation and Legendre polynomials $P_n(x)$ and $Q_n(x)$ , Generating function for $P_n(x)$ (derivation), Rodrigue's formula, Various recurrence relations (derivations), Orthogonality of Legendre Polynomials. | .      |

#### Reference Books:

1. Topics in Algebra : I. N. Herstein, Wiley, 2006.
2. A First course in Abstract Algebra : Fraleigh J.B, Pearson, 2013.
3. Elements of Modern Algebra and Topology : Sampathkumar & K. S. Amur.
4. Modern Algebra : P.N Chatterjee, Anu Books, 2019.
5. Introduction to Mathematical Analysis: Shanti Narayan ( S.Chand & Co.)
6. A Course in Abstract Algebra : Vijay K Khanna and Bambari, Vikash Publisher, 2017.
7. Modern Algebra : M. L Khanna, Jai Prakash Nath and Co., 2012.
8. A Course in B.Sc. Mathematics : Bhoosnurmath, C S Salimath & V S Shetiya
9. College Mathematics for B.Sc.: N. Rudraiah
10. A Text Book of B.Sc. Mathematics : G K Ranganath
11. Special Functions: G. E. Andrews, R. Askey and R. Roy, Cambridge University Press
12. Vistas of special functions: S. Kanemitsu and H. Tsukada, World Scientific 2010
13. Thomas Calculus: G. B. Thomas, Pearson publication, 13<sup>th</sup> Edition, 2014
14. Higher Engineering Mathematics: B. S. Grewal, Khanna Publications,
15. Mathematical Methods for Physics and Engineering,: K. F. Riley, M. P. Hobson and S. J. Bence, Third Edition, Cambridge University Press, 3<sup>rd</sup> Edition 2006.

| Formative Assessment for Theory        |                |
|----------------------------------------|----------------|
| Assessment Occasion / Type             | Marks          |
| Internal Assessment Test-1             | 05             |
| Internal Assessment Test-2             | 05             |
| Assignment                             | 10             |
| <b>Total</b>                           | <b>20Marks</b> |
| Formative Assessment as per guidelines |                |

**B.Sc. Semester – V**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Elective (DSE)**  
**Course Title: Linear Algebra and Analytical Geometry**  
**Course Code: C5MAT2T2 (Theory)**

**SPECIALIZATION – II**  
**(Student shall select DSE – 1 (T6) or DSE – 2 (T7) for 03 credits only)**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSE – 2 (T7)   | Theory            | 03      | 04                         | 60 Hrs.                                    | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** After completion of the course (Theory), students will be able to:

- CO1:** Understand vector spaces, subspaces, and linear independence.
- CO2:** Apply linear transformations, matrices, and eigenvalues techniques.
- CO3:** Analyze the properties and equations of spheres.
- CO4:** Solve problems involving cones and cylinders.

| Unit            | Linear Algebra and Analytical Geometry                                                                                                                                                                                | 60 Hours / Sem |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Unit-I</b>   | <b>Vector Space:</b>                                                                                                                                                                                                  | 15 Hrs.        |
|                 | Introduction to Rings, Fields and Vector space with examples, properties, Vector, Subspace. Linear independence and dependence of vectors. Linear span of a set. Base for vector space and dimension.                 |                |
| <b>Unit-II</b>  | <b>Linear transformation:</b>                                                                                                                                                                                         | 15 Hrs.        |
|                 | Definition and examples, properties, range, null space, rank and nullity theorem, Linear maps as matrices-change of basis and effect of associated matrices, eigenvalues and eigenvectors of a linear transformation. |                |
| <b>Unit-III</b> | <b>Sphere:</b>                                                                                                                                                                                                        | 15 Hrs.        |

|                |                                                                                                                                                                                                                                                     |         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|                | Equation of a Sphere. Equation of a sphere with two given points as the ends of diameter (Different forms), Intersection of two spheres, Section of a sphere by a plane. Equation of a Sphere through a circle. Tangent planes. Orthogonal spheres. |         |
| <b>Unit-IV</b> | <b>Cone and Cylinder:</b>                                                                                                                                                                                                                           | 15 Hrs. |
|                | Equation of a cone, Quadric cone. Enveloping cone of a sphere, Right circular cone, Equation of a cylinder, Enveloping cylinder of a sphere, and Right circular cylinder.                                                                           |         |

### Recommended Books:

1. Elements of Analytical Solid Geometry, Shanti Narayan and P. K. Mittal, S. Chand & Co., 2007.
2. Analytical Geometry, A. R. Vasishtha and D. C. Agarwal, Krishna Prakashan Media, 2014.
3. Analytical Solid Geometry, B. P. Singh, Venus Books, 2015.
4. Mathematical Analysis: Shanti Narayan , 2013
5. College Mathematics for B.Sc, N Rudraiah and others.
6. First course in Real Analysis, M.K. Singhal and Asha Rani. 2003
7. Modern Algebra: M.L. Khanna, Jai Prakash Nath & Co. 2012
8. A Course in Abstract Algebra: Vijay K Khanna and S K Bhambri, Pearson 2013
9. Elements of Modern Algebra and Topology, Sampathkumar & K. S. Amur, 1994

| <b>Formative Assessment for Theory</b>        |                |
|-----------------------------------------------|----------------|
| <b>Assessment Occasion / Type</b>             | <b>Marks</b>   |
| Internal Assessment Test-1                    | 05             |
| Internal Assessment Test-2                    | 05             |
| Assignment                                    | 10             |
| <b>Total</b>                                  | <b>20Marks</b> |
| <b>Formative Assessment as per guidelines</b> |                |

**B.Sc. Semester – VI**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Course (DSC)**  
**Course Title: Numerical Analysis**  
**Course Code: C6MAT1T1 (Theory)**  
**Compulsory Theory paper**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC – 6 (T8)   | Theory            | 03      | 04                         | 60 Hrs.                                    | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** After completion of the course (Theory), students will be able to:

- CO 1:** Understand the concept of errors.
- CO 2:** Find roots of equations using various iterative methods.
- CO 3:** Analyse system of linear equations using direct and iterative methods.
- CO 4:** Describe various operators such as difference operators, shift operators and so on.
- CO 5:** Apply numerical techniques in other areas of mathematics.
- CO 6:** Understand the application of finite differences in finding derivatives.
- CO 7:** Find the value of definite integrals using numerical methods.

| Unit      | Title: Numerical Analysis                                                                                                                                                                                                                                                 | 60 Hours / Sem |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Unit - I  | <b>Algebraic and Transcendental Equations:</b>                                                                                                                                                                                                                            | <b>15 Hrs.</b> |
|           | Errors: Significant digits, Absolute, Relative, Percentage errors, Rounding off and Truncation errors, Solutions of algebraic and transcendental equations: Bisection method, Regula-falsi method, Secant method, Newton-Raphson method and Fixed-point iterative method. |                |
| Unit - II | <b>System of Linear Algebraic Equations:</b>                                                                                                                                                                                                                              | <b>15 Hrs.</b> |

|                   |                                                                                                                                                                                                                                                                                                |                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|                   | Direct Methods: Gauss elimination method, Gauss-Jordan method and Triangularization method. Iterative methods: Gauss-Jacobi method, Gauss-Seidel method, Successive – Over Relaxation (SOR) method.                                                                                            |                |
| <b>Unit – III</b> | <b>Finite Differences:</b>                                                                                                                                                                                                                                                                     | <b>15 Hrs.</b> |
|                   | Finite differences, Forward, Backward and Shift Operators: Definitions, Properties and Problems; Polynomial Interpolation: Newton – Gregory forward and backward interpolation formulas, Lagrange’s interpolation formula and problems.                                                        |                |
| <b>Unit - IV</b>  | <b>Numerical Differentiation and Integration:</b>                                                                                                                                                                                                                                              | <b>15 Hrs.</b> |
|                   | Formula for derivatives (first and second order) based on Newton-Gregory forward and backward interpolations, General quadrature formula, Trapezoidal rule, Simpson’s (1/3) <sup>rd</sup> rule, Simpson’s (3/8) <sup>th</sup> rule (derivations), Weddle’s rule (Statement only) and problems. |                |

### Recommended Books:

1. Analysis of Numerical Methods, E.Isaacson and H.B.Keller,Dover Publications, 2012.
2. Introductory Methods of Numerical Analysis, S.S.Sastry, PHI Learning Pvt.Ltd., 2012.
3. Advanced Engineering Mathematics, E. Kreyszig,Wiley, 2006.
4. Numerical Methods for Scientists and Engineers, K. Sankara Rao, PHI Learning, 2018.
5. Numerical Methods for Scientists and Engineers, R. W. Hamming, Dover Publications, 1987.
6. Numerical Methods for Scientists and Engineers, J. D. Hoffman, CRC Press, 2001.
7. Numerical Methods in Engineering and Science, B.S.Grewal, Khanna Publishers, 2013.
8. Numerical Methods for Scientific and Engineering Computation, M.K.Jain, S.R.K. Iyengar and R K.Jain, New Age International Publishers, 2022.
9. Finite Differences and Numerical Analysis, H.C.Saxena, S.Chand Publication, 2010.
10. Numerical Analysis, B.D. Gupta, Konark Publishers Pvt.Ltd., 1990.
11. A Course in B.Sc. Mathematics-V, S. S. Bhoosnurmath, C.S. Salimath and V. S. Shetiya, EBPB, 2018.

| <b>Formative Assessment for Theory</b>        |                |
|-----------------------------------------------|----------------|
| <b>Assessment Occasion / Type</b>             | <b>Marks</b>   |
| Internal Assessment Test-1                    | 05             |
| Internal Assessment Test-2                    | 05             |
| Assignment                                    | 10             |
| <b>Total</b>                                  | <b>20Marks</b> |
| <b>Formative Assessment as per guidelines</b> |                |

**B.Sc. Semester – VI**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Course (DSC)**  
**Course Title: Practical on Numerical Analysis**  
**Course Code: C6MAT1P1 (Practical)**  
**Compulsory Practical**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSC – 6 (P6)   | Practical         | 02      | 04                         | 56 Hrs.                                    | 03 Hrs.                  | 10                         | 40                         | 50          |

**Course Outcomes (COs):** After completion of course (Practical), students will be able to:

- CO 1:** Learn Free and Open Source software (FOSS) tools for computer programming.
- CO 2:** Solve problems on Numerical Analysis using Python language.
- CO 3:** Acquire knowledge of applications of numerical analysis.
- CO 4:** Find roots of equations using various numerical techniques.
- CO 5:** Solve system of equations using direct and iterative methods.
- CO 6:** Find the derivative and integral of a function using numerical methods.

**List of Experiments**

**[Each will have 4 Hrs. / Week (Minimum 12 experiments)]**

1. Program to find the root of an equation using Bisection and Regula-Falsi methods.
2. Program to find the root of an equation using Newton-Raphson and Secant methods.
3. Program to solve the system of algebraic equations using the Gauss-elimination method.
4. Program to solve the system of algebraic equations using the Gauss-Jordan method.
5. Program to solve the system of algebraic equations using the Gauss-Jacobi method.
6. Program to solve the system of algebraic equations using the Gauss-Seidel method.
7. Program to find the value of  $y = f(x)$  for the given data using Newton–Gregory technique.
8. Program to find the polynomial equation from the given data using the Lagrange’s interpolation formula for unequal intervals.
9. Program to find derivative of  $y = f(x)$  at specified point using the Newton Gregory interpolation method
10. Program to evaluate integral using Trapezoidal rule.
11. Program to evaluate integral using Simpson’s 1/3 rule.
12. Program to evaluate integral using Simpson’s 3/8 rule.

**Recommended Books:**

1. Automate the Boring Stuff with Python, A. Sweigart, No Starch Press, 2019.
2. Python Cookbook, D. Beazley and B. K. Jones, Shroff O'Reilly, 2013.
3. Basic Python Programming for Beginners, K. V. Rajkumar, M. Krishna, and J. Prakash, Bluerose Publishers Pvt. Ltd., 2021.
4. Python, J. Shovic and A. Simpson, For Dummies, 2024.
5. Learning Python, M. Lutz, Shroff O'Reilly, 2017.
6. Analysis of Numerical Methods, E. Isaacs on and H.B.Keller, Dover Publications, 2012.
7. Introductory Methods of Numerical Analysis, S.S.Sastry, PHI Learning Pvt. Ltd., 2012.
8. Advanced Engineering Mathematics: E. Kreyszig, Wiley, 2006.
9. Numerical Methods for Scientists and Engineers, K. Sankara Rao, PHI Learning, 2018.
10. Finite Differences and Numerical Analysis, H.C.Saxena, S.Chand Publication, 2010.

**General Instructions****Software to be used: PYTHON****Scheme of Practical Examination (Distribution of Marks)**

| <b>Formative Assessment for Practical</b>       |                |
|-------------------------------------------------|----------------|
| <b>Assessment Occasion /Type</b>                | <b>Marks</b>   |
| Program writing and problem solving (1 Program) | 03             |
| Program Execution                               | 03             |
| Viva                                            | 02             |
| Journal                                         | 02             |
| <b>Total</b>                                    | <b>10Marks</b> |
| <b>Formative Assessment as per guidelines.</b>  |                |

| <b>Summative Assessment for Practical</b>        |                |
|--------------------------------------------------|----------------|
| <b>Assessment Occasion /Type</b>                 | <b>Marks</b>   |
| Program writing and problem solving (2 Programs) | $(5+5)*2 = 20$ |
| Program Execution (Any 1 Program)                | 10             |
| Viva                                             | 05             |
| Journal                                          | 05             |
| <b>Total</b>                                     | <b>40Marks</b> |
| <b>Summative Assessment as per guidelines.</b>   |                |

B.Sc. Semester – VI

**Subject: MATHEMATICS**

**Discipline Specific Core Elective (DSE)**

**Course Title: Complex Analysis and Topology**

**Course Code: C6MAT2T1 (Theory)**

**SPECIALIZATION – I**

**(Student shall select DSE – 3 (T9) or DSE – 4 (T10) for 03 credits only)**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSE – 3 (T9)   | Theory            | 03      | 04                         | 60 Hrs.                                    | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** After completion of the course (Theory), students will be able to:

- CO 1:** Understand the complex numbers and the concepts of analytic, harmonic and integral functions
- CO 2:** Verify Cauchy-Riemann equations in Cartesian and polar forms.
- CO 3:** Construct analytic functions using Milne-Thomson's method.
- CO 4:** Solve complex integrals using Residue Theorem
- CO 5:** Learn the concepts of closure, interior and boundary of a set.
- CO 6:** Understand the concepts of basis and sub-basis.
- CO 7:** Analyse  $T_0, T_1$  and  $T_2$  spaces.

| Unit    | Title: Complex Analysis and Topology                                                                                                                                                                                                                                                                                                                                                                                                                                           | 60 Hours /Sem |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Unit-I  | <b>Complex Numbers:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 15 Hrs        |
|         | Complex numbers in polar form; De Moivre's Theorem (statement only); $n^{\text{th}}$ roots of a complex number; expansion of $\sin n\theta, \cos n\theta$ and $\tan n\theta$ in terms of $\sin \theta, \cos \theta$ , and $\tan \theta$ respectively; expansion of $\sin^n \theta, \cos^n \theta$ , and $\sin^m \theta \cos^n \theta$ as a series of sines or cosines of multiples of $\theta$ ; logarithm of complex numbers; summation of trigonometric series in C+iS form. |               |
| Unit-II | <b>Functions of a Complex Variable:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        | 15 Hrs        |

|                 |                                                                                                                                                                                                                                                                                                                             |        |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|                 | Limit, continuity, and differentiability of a complex function; Analytic functions; Cauchy–Riemann equations in Cartesian and polar forms; sufficient conditions for analyticity (Cartesian form); harmonic functions; properties of analytic functions; construction of analytic functions using the Milne–Thomson method. |        |
| <b>Unit-III</b> | <b>Complex Integration:</b>                                                                                                                                                                                                                                                                                                 | 15 Hrs |
|                 | Complex line integrals; Cauchy's Integral Theorem; Taylor's series; Laurent's series; Residue Theorem; contour integration and related problems.                                                                                                                                                                            |        |
| <b>Unit-IV</b>  | <b>Topology:</b>                                                                                                                                                                                                                                                                                                            |        |
|                 | Definition of topological space and examples, Closure, Neighbourhood, Limit points and Derived sets, Interior, Exterior and Boundary, Bases and sub-bases, Subspaces, $T_0$ , $T_1$ and $T_2$ spaces                                                                                                                        | 15 Hrs |

#### Recommended Books:

1. Complex Variables, J.N. Sharma, Krishna Prakashan, 2014.
2. Complex Analysis, Goyal, Gupta and Pundir, Anu Books, 2019.
3. Complex Variables, Spiegel and Others, Schaum's Outline Series, 2009.
4. Complex Analysis, L. V. Ahlfors, McGraw Hill Education, 2017.
5. Complex Analysis, S. Lang, Springer, 2003.
6. Theory of Functions of a Complex Variable, Shanthi Narayan and P. K. Mittal, S.Chand Publishers, 2005.
7. Foundations of Complex Analysis, S. Ponnusamy, Narosa Publications, 2011.
8. A Course in B.Sc. Mathematics – VI, S. S. Bhoosnurmath, C. S. Salimath and V.S. Shetiya, EBPB, 2018
9. Topology, J.R. Munkres, Pearson, 2021.
10. Topology, J.N. Sharma and J.P. Chauhan, Krishna Prakashan, Meerut, 2014.
11. General Topology, S. Lipschutz, Schaum's Outlines Series, 1965.
12. Elements of Modern Algebra and Topology, E. Sampathkumar and K.S. Amur, Vidya Vikas Publications, 1969.
13. General Topology, S.R. Malghan, Serial Publication, 2013.

| Formative Assessment for Theory        |                |
|----------------------------------------|----------------|
| Assessment Occasion / Type             | Marks          |
| Internal Assessment Test-1             | 05             |
| Internal Assessment Test-2             | 05             |
| Assignment                             | 10             |
| <b>Total</b>                           | <b>20Marks</b> |
| Formative Assessment as per guidelines |                |

**B.Sc. Semester – VI**  
**Subject: MATHEMATICS**  
**Discipline Specific Core Elective (DSE)**  
**Course Title: Mechanics and Mathematical Modelling**  
**Course Code: C6MAT2T2 (Theory)**

**SPECIALIZATION – II**

(Student shall select DSE –1 (T6) OR DSE-2 (T7) for 03 credits only)

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| DSE – 4 (T10)  | Theory            | 03      | 04                         | 60 Hrs.                                    | 03 Hrs.                  | 20                         | 80                         | 100         |

**Course Outcomes (COs):** After completion of the course (Theory), students will be able to:

- CO 1:** Analyse the concepts of couple and moment of couple.
- CO 2:** Solve problems on Radial and Transverse velocity as well as acceleration.
- CO 3:** Understand the concepts of direct and oblique impacts.
- CO 4:** Evaluate problems related to tangential and normal components of acceleration.
- CO 5:** Understand the need for Mathematical Modelling.
- CO 6:** Model problems in Algebra, Geometry, etc.
- CO 7:** Prepare a variety of growth and decay models.

| Unit             | Title: Mechanics and Mathematical Modelling                                                                                                                                                                                           | 60 Hours / Sem |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Unit – I</b>  | <b>Statics:</b>                                                                                                                                                                                                                       | 15 Hrs.        |
|                  | Couples, moment of a couple, Varignon's theorem. Resultant of coplanar couples. Resultant of a force and a couple. Resultant of a system of coplanar forces acting at different points of a rigid body. Catenary and common catenary. |                |
| <b>Unit – II</b> | <b>Dynamics – I:</b>                                                                                                                                                                                                                  | 15 Hrs.        |

|                   |                                                                                                                                                                                                                                                                                                                                                                                      |         |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|                   | Velocity and acceleration of a particle along a plane curve. Radial and transverse components of velocity and acceleration. Tangential and Normal components of velocity and acceleration.                                                                                                                                                                                           |         |
| <b>Unit – III</b> | <b>Dynamics – II:</b>                                                                                                                                                                                                                                                                                                                                                                | 15 Hrs. |
|                   | Motion of a projectile in a non-resisting medium under gravity. Motion of a particle under a central force. Use of polar co-ordinates and pedal coordinates. Elastic impact-Direct and oblique impact of elastic bodies.                                                                                                                                                             |         |
|                   | <b>Mathematical Modelling:</b>                                                                                                                                                                                                                                                                                                                                                       | 15 Hrs. |
| <b>Unit – IV</b>  | Need for mathematical Modelling, Brachistochrone problem with historical developments. Techniques and classification of mathematical models. Mathematical Modelling through Geometry, Algebra (Compound Interest Model), Trigonometry and Calculus. Mathematical Modelling through differential equations of first order. Growth and Decay models (linear). Prey and predator model. |         |

#### Recommended Books:

1. Statics, B. C. Das and B. N. Mukherjee, U. N. Dhar & Sons Pvt. Ltd., 2018.
2. Statics, H. Kishan, Atlantic Publishers, 2023.
3. Engineering Mechanics Statics, J. L. Meriam, L. G. Kraige and J. N. Bolton, Wiley, 2017.
4. The Elements of Statics and Dynamics, S. L. Loney, Arihant Publication, 2023.
5. Dynamics, H. Kishan, Atlantic Publishers, 2023.
6. Engineering Dynamics, M. R. Islam, A. K. M. Mazumder and M. Ahmed, CRC Press, 2022.
7. Engineering Mechanics, I. H. Shames and G. K. Mohana Rao, Pearson, 2005.
8. A Textbook on Dynamics, M. Ray and G. C. Sharma, S. Chand & Company, 2005.
9. Mechanics, B. Singh and S. K. Pundir, Pragati Prakashan, 2020.
10. Engineering Mechanics, A. Nelson, McGraw Hill Edu., 2017.
11. Mathematical Modelling, J.N. Kapur, New Age International Publishers, 2023.
12. Mathematical Modelling Techniques, R. Aris, Dover Publications, 2003.
13. Fundamental Mathematical Modelling, S. Jana, S. Rayachaudhuri and B. Pal, Santra Publication Pvt. Ltd., 2022.

| Formative Assessment for Theory               |                |
|-----------------------------------------------|----------------|
| Assessment Occasion / Type                    | Marks          |
| Internal Assessment Test-1                    | 05             |
| Internal Assessment Test-2                    | 05             |
| Assignment                                    | 10             |
| <b>Total</b>                                  | <b>20Marks</b> |
| <b>Formative Assessment as per guidelines</b> |                |

## B.Sc. Semesters – IV/V/VI

### Subject: MATHEMATICS

#### SKILL ENHANCEMENT COURSE (SEC)

(Student shall study this SEC in any one of the Semesters either in IV or V or VI semester and College shall decide to allot the students)

**Course Title: Programming with GNU Octave**

**Course Code: C0MAT6P1 (Practical)**

| Type of Course | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|----------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| SEC            | Practical         | 02      | 04                         | 56 Hrs.                                    | 03 Hrs.                  | 10                         | 40                         | 50          |

**Course Outcomes (COs):** After completion of course (Practical), students will be able to:

**CO 1:** Learn GNU Octave Software (FOSS) for computer programming.

**CO 2:** Solve problems on Algebra, Calculus, Differential Equations, Laplace Transforms and Fourier series by using Octave software.

**CO 3:** Acquire knowledge of applications of abovementioned topics through Octave.

**CO 4:** Find angle between radius vector and tangent using FOSS.

**CO 5:** Verify the exactness of a differential equation.

**CO 6:** Calculate double integral with constant and variable limits.

#### List of Experiments

**[Each will have 4 Hrs. / Week (Minimum 12 experiments)]**

1. Program to compute addition and subtraction of matrices
2. Program to compute rank of matrix and row reduced echelon form.
3. Program to find the angle between the radius vector and tangent.
4. Program to find the  $n^{\text{th}}$  derivative of  $e^{ax}$  and trigonometric functions.
5. Program to construct Cayley table and test abelian for given finite set.
6. Programs to verify Lagrange's theorem (groups) with suitable example.
7. Program to evaluate the double and triple integrals with constant limits.
8. Program to verify the exactness of a differential equation.
9. Program to find the Complementary Function and Particular Integral of linear differential equations with constant coefficients.
10. Program to verify D'Alembert's and Raabe's tests with suitable examples.
11. Program to find the Laplace transforms of some standard and periodic functions.
12. Program to find sine and cosine Fourier transforms.

**Recommended Books:**

1. Introduction to Octave, N. Sandeep, APress, 2017.
2. Scientific Computing with Matlab and Octave, A. Quarteroni, Springer, 2010.
3. Essential Matlab and Octave, J. Rogel-Salazar, CRC Press, 2018.
4. GNU Octave by Example, A. Pajankar, APress, 2020.
5. Octave and Matlab for Engineering Applications, A. Stahel, Springer, 2022.
6. Differential Calculus, Shanti Narayan and P. K. Mittal, S. Chand & Company, New Delhi, 2022.
7. Modern Algebra, S. Singh and Q. Zameeruddin, Vikas Publishing House Pvt. Ltd., 2006.
8. Ordinary Differential Equations and Partial Differential Equations, M. D. Raisinghania, S. Chand & Company, New Delhi, 2020.
9. Laplace Transforms, M. R. Spiegel (Schaum's Series), McGraw-Hill International Ed., 2005.
10. Fourier Series and Integral Transforms, S. Sreenadh & Others, S. Chand & Company, 2014.

**General Instructions****Software to be used: GNU OCTAVE****Scheme of Practical Examination (Distribution of Marks)**

| <b>Formative Assessment for Practical</b>       |                |
|-------------------------------------------------|----------------|
| <b>Assessment Occasion /Type</b>                | <b>Marks</b>   |
| Program writing and problem solving (1 Program) | 03             |
| Program Execution                               | 03             |
| Viva                                            | 02             |
| Journal                                         | 02             |
| <b>Total</b>                                    | <b>10Marks</b> |
| <i>Formative Assessment as per guidelines.</i>  |                |

| <b>Summative Assessment for Practical</b>        |                |
|--------------------------------------------------|----------------|
| <b>Assessment Occasion /Type</b>                 | <b>Marks</b>   |
| Program writing and problem solving (2 Programs) | $(5+5)*2 = 20$ |
| Program Execution (Any 1 Program)                | 10             |
| Viva                                             | 05             |
| Journal                                          | 05             |
| <b>Total</b>                                     | <b>40Marks</b> |
| <i>Summative Assessment as per guidelines.</i>   |                |

**INTERNSHIP**  
**Course Title: Project Work / Internship**

**Course Code: C6MAT7PR / C6MAT7IN**

| Type of Course           | Theory /Practical | Credits | Instruction hours per week | Total No. of Lectures / Hours per Semester | Duration of Sem End Exam | Formative Assessment Marks | Summative Assessment Marks | Total Marks |
|--------------------------|-------------------|---------|----------------------------|--------------------------------------------|--------------------------|----------------------------|----------------------------|-------------|
| Project Work/ Internship | Practical         | 02      | 04                         | 56 Hrs.                                    | 03 Hrs.                  | 10                         | 40                         | 50          |

**Course Outcomes (COs): At the end of the course students will be able to:**

- CO1:** Identify and formulate mathematical problems or real-world applications suitable for project work or internship.
- CO2:** Apply mathematical concepts, computational techniques, Artificial Intelligence (AI) tools, and appropriate software to solve project or internship-related problems.
- CO3:** Develop, analyse, and validate mathematical models, algorithms, computational methods, or data-driven solutions.
- CO4:** Interpret and evaluate the results obtained through project work or internship using mathematical reasoning and computational techniques.
- CO5:** Prepare a professional project/internship report and effectively present the outcomes through seminars, demonstrations, and viva-voce.
- CO6:** Demonstrate professional ethics, teamwork, communication, leadership, and time-management skills while carrying out project or internship activities.
- CO7:** Integrate mathematical knowledge, computational skills, AI tools, and practical experience to address real-world, industrial, research, or societal problems.

**A. Guidelines for Project Work Using Artificial Intelligence (AI)**

1. The Project Work shall be carried out under the guidance of a faculty member approved by the Department, using appropriate Artificial Intelligence (AI) tools.
2. The project may be undertaken within the college or at recognized educational institutions, universities, research laboratories, R & D organizations, industries, or other approved organizations, subject to the prior approval of the competent authority.
3. The project may involve one or more of the following:
  - Literature survey and review using AI-assisted research tools.
  - Mathematical modelling and problem-solving.
  - Development of mathematical algorithms and computational methods.
  - AI-assisted programming, data analysis, visualization, or simulation.
  - Applications of Artificial Intelligence in Mathematics and interdisciplinary domains.

- Preparation of educational content, technical reports, presentations, and mathematical documentation using AI tools with proper verification and appropriate citation, wherever applicable.
- 4. Students are encouraged to use suitable AI tools responsibly for ideation, literature review, coding assistance, data analysis, visualization, documentation, and presentation. The final project report and all submitted work shall reflect the student's own understanding, analysis, interpretation, and original contribution.
- 5. The project may be carried out individually or in groups, subject to the approval of the Department.
- 6. Students shall maintain a project diary/logbook documenting the progress of the work and submit a final project report in the prescribed format.
- 7. At the completion of the project, students shall present their work through a seminar and appear for a viva-voce examination before the evaluation committee.
- 8. The evaluation shall be based on the quality of the project work, effective use of appropriate mathematical techniques and AI tools, originality, report writing, seminar presentation, and performance in the viva-voce examination.

## **B. Guidelines for Internship**

1. Internship is a discipline-specific experiential learning activity carried out under the guidance of an external mentor/supervisor and a faculty mentor.
2. The internship may be undertaken in educational institutions, universities, research laboratories, R&D organizations, industries, government organizations, local bodies, business organizations, start-ups, NGOs, IT companies, financial institutions, or any other organization approved by the Department.
3. The internship shall be of **45–60 hours (2 Credits)**. It may be undertaken either:
  - on a **full-time basis** for **1–2 weeks** immediately before the closure of the semester, or
  - on a **part-time basis** for **4 hours per week** during the academic session for **13–14 weeks**.
 The mode of implementation shall be decided by the college programme-wise and shall not be determined subject-wise.
4. One credit of internship shall be equivalent to **30 hours** of field/workplace engagement.
5. The faculty mentor shall guide, monitor, and evaluate the progress of the internship. The mentor's workload shall be as per the University norms.
6. Every student shall maintain an internship diary/logbook and submit an internship report in the prescribed format, duly certified by the organization and the faculty mentor.
7. At the completion of the internship, the student shall deliver a seminar presentation and appear for a viva-voce examination.
8. Evaluation shall be based on:
  - Performance during the internship.
  - Internship diary/logbook and final report.
  - Seminar presentation.
  - Viva-voce examination.
9. Students shall maintain professional ethics, discipline, punctuality, and confidentiality during the internship and comply with the rules and regulations of the host organization.

| <b>Formative Assessment for Practical</b>             |                |
|-------------------------------------------------------|----------------|
| <b>Assessment Occasion /Type</b>                      | <b>Marks</b>   |
| Progress Report-1                                     | 05             |
| Progress Report-2                                     | 05             |
| <b>Total</b>                                          | <b>10Marks</b> |
| <b><i>Formative Assessment as per guidelines.</i></b> |                |

| <b>Summative Assessment for Practical</b>             |                |
|-------------------------------------------------------|----------------|
| <b>Assessment Occasion /Type</b>                      | <b>Marks</b>   |
| Project/Internship Report                             | 20             |
| Presentation                                          | 15             |
| Viva                                                  | 05             |
| <b>Total</b>                                          | <b>40Marks</b> |
| <b><i>Summative Assessment as per guidelines.</i></b> |                |

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**B.Sc. Programme: 2026-27**  
**GENERAL PATTERN OF THEORY QUESTION PAPER**  
**FOR DSC/DSE**  
**(80 marks for Semester-End Examination with 3 Hrs duration)**

Part-A

1. Question Numbers 1-10 carry 2 marks each. : 20 marks

Part-B

2. Question Numbers 11- 18 carry 05 Marks each. Answer any 06 questions : 30 marks

Part-C

3. Question Numbers 19-22 carry 10 Marks each. Answer any 03 questions : 30 marks  
(Minimum 1 question from each unit and 10 marks question may have sub-questions  
for 7+3 or 6+4 or 5+5 if necessary)

**Total: 80 Marks**

**Note:** Proportionate weightage shall be given to each unit based on number of hours prescribed

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